Differentiation

Definition: The Rate of change of dependent variable with respect to independent variable is called derivative and The process of finding derivative is called Differentiation.

Q 1: Find the first derivative of the following function

(a) $\frac{3x^2 + 5x}{7x + 4}$ (b) $\frac{(2x + 1)(3x + 1)}{4x + 1}$

Sol:(a)

let y $= \frac{3x^2 + 5x}{7x + 4}$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx}\left(\frac{3x^2 + 5x}{7x + 4}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{(7x + 4)\frac{d}{dx}(3x^2 + 5x) - (3x^2 + 5x)\frac{d}{dx}(7x + 4)}{(7x + 4)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(7x + 4)(6x + 5) - (3x^2 + 5x) \cdot 7}{(7x + 4)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{42x^2 + 35x + 24x + 20 - 21x^2 - 35x}{(7x + 4)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{21x^2 + 24x + 20}{(7x + 4)^2}$$

(solved)

Sol:(b)

$$\frac{(2x + 1)(3x + 1)}{4x + 1}$$

$$= \frac{6x^2 + 2x + 3x + 1}{4x + 1}$$

$$= \frac{6x^2 + 5x + 1}{4x + 1}$$

let, y $= \frac{6x^2 + 5x + 1}{4x + 1}$

$\Rightarrow \frac{dy}{dx} = \frac{d}{dx}\frac{6x^2 + 5x + 1}{4x + 1}$

$\Rightarrow \frac{dy}{dx} = \frac{(4x+1)\frac{d}{dx}(6x^2 + 5x + 1) - (6x^2 + 5x + 1)\frac{d}{dx}(4x+1)}{(4x+1)^2}$

$\Rightarrow \frac{dy}{dx} = \frac{(4x+1)(12x+5) - (6x^2 + 5x + 1) \cdot 4}{(4x+1)^2}$

$\Rightarrow \frac{dy}{dx} = \frac{(4x+1)(12x+5) - (6x^2 + 5x + 1) \cdot 4}{(4x+1)^2}$

$\Rightarrow \frac{dy}{dx} = \frac{24x^2 + 12x + 1}{(4x+1)^2}$

(solved)

Q 2: Find the first derivative of the following function

(a) $3\sqrt{x}tanx$ (b) x^3tanx

Sol:(a)

let y=$3\sqrt{x}tanx$

$\Rightarrow \frac{dy}{dx} = 3\frac{d}{dx}(\sqrt{x}tanx)$

$\Rightarrow \frac{dy}{dx} = 3[\sqrt{x}\frac{d}{dx}tanx + tanx\frac{d}{dx}\sqrt{x}]$

$\Rightarrow \frac{dy}{dx} = 3[\sqrt{x}\frac{d}{dx}sec^2x + tanx \cdot \frac{1}{2\sqrt{x}}]$

$\Rightarrow \frac{dy}{dx} = 3\sqrt{x}sec^2x + \frac{tanx}{2\sqrt{x}}$

(solved)

Sol:(b)

let y $=x^3tanx$

$$\Rightarrow \frac{dy}{dx} = x^3 \frac{d}{dx} tanx + tanx \frac{d}{dx} x^3$$
$$\Rightarrow \frac{dy}{dx} = x^3 sec^2 x + tanx \cdot 3x^2$$
$$\Rightarrow \frac{dy}{dx} = x^3 sec^2 x + 3x^2 \cdot tanx$$

(solved)

Q 3: If $y = \sqrt{\frac{1-x}{1+x}}$ then prove that $(1-x)^2 \frac{dy}{dx} + y = 0$

Sol:

Given that,

$$y = \sqrt{\frac{1-x}{1+x}}$$
$$\Rightarrow y = \sqrt{\frac{(1-x)(1-x)}{(1+x)(1-x)}}$$
$$\Rightarrow y = \sqrt{\frac{(1-x)^2}{1-x^2}}$$
$$\Rightarrow y = \frac{1-x}{\sqrt{1-x^2}} \text{..............}(i)$$
$$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{1-x^2}\frac{d}{dx}(1-x) - (1-x)\frac{d}{dx}\left(\sqrt{1-x^2}\right)}{(\sqrt{1-x})^2}$$
$$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{1-x^2}(-1) - (1-x)\frac{1}{2\sqrt{1-x^2}}\frac{d}{dx}(1-x^2)}{1-x^2}$$
$$\Rightarrow \frac{dy}{dx} = \frac{-\sqrt{1-x^2} - \frac{1-x}{2\sqrt{1-x^2}}(-2x)}{1-x^2}$$
$$\Rightarrow \frac{dy}{dx} = \frac{-\sqrt{1-x^2} + \frac{(1-x)x}{\sqrt{1-x^2}}}{1-x^2}$$
$$\Rightarrow \frac{dy}{dx} = \frac{-\sqrt{1-x^2} + \frac{x-x^2}{\sqrt{1-x^2}}}{1-x^2}$$
$$\Rightarrow (1-x^2)\frac{dy}{dx} = \frac{-\sqrt{1-x^2}^2 + x - x^2}{\sqrt{1-x^2}}$$
$$\Rightarrow (1-x^2)\frac{dy}{dx} = \frac{-1+x^2+x-x^2}{\sqrt{1-x^2}}$$
$$\Rightarrow (1-x^2)\frac{dy}{dx} = \frac{-(1-x)}{\sqrt{1-x^2}}$$

$\Rightarrow (1-x^2)\frac{dy}{dx} = -y$ ....from (i)

$$\Rightarrow (1-x^2)\frac{dy}{dx} + y = 0$$

(proved)

Q 4: If $siny = xsin(a+y)$ then show that $\frac{dy}{dx} = \frac{sin^2(a+y)}{sina}$

Sol:

Given that,

$$siny = xsin(a+y)$$

$$\Rightarrow x = \frac{siny}{sin(a+y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{sin(a+y)\frac{d}{dy}siny \frac{d}{dy}sin(a+y)}{sin^2(a+y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{sin(a+y)cosy - sinycos(a+y)\frac{d}{dy}(a+y)}{sin^2(a+y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{sin(a+y)cosy - cos(a+y)siny}{sin^y(a+y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{sin(a+y-y)}{sin^2(a+y)}$$ [sin(A-B)=sinA cosB - cosA sinB]

$$\Rightarrow \frac{dy}{dx} = \frac{sin^2(a+y)}{sina}$$

(solved)

Q 5: If $y = x^3log(\frac{1}{x})$ then prove that $x\frac{dy^2}{dx^2} - 2\frac{dy}{dx} + 3x^2 = 0$

Sol:

Given that,

$y = x^3log(\frac{1}{x})$ ………..(i)

differentiating with respect to x we get,

$\Rightarrow \frac{dy}{dx} = x^3 \frac{d}{dx} log(\frac{1}{x}) + log \frac{1}{x} \frac{d}{dx} x^3$

$\Rightarrow \frac{dy}{dx} = x^3 \frac{1}{\frac{1}{x}} \frac{d}{dx} \frac{1}{x} + log \frac{1}{x} \cdot 3x^2$

$\Rightarrow \frac{dy}{dx} = x^4 (-\frac{1}{x^2}) + 3x^2 log \frac{1}{x}$

$\Rightarrow \frac{dy}{dx} = - x^2 + 3x^2 log \frac{1}{x}$

$\Rightarrow x \frac{dy}{dx} = - x^3 + 3x^3 log \frac{1}{x}$……….[by multiplying x]

$\Rightarrow x \frac{dy}{dx} = - x^3 + 3y$……….from (I)

$\Rightarrow x \frac{d^2y}{dx^2} + \frac{dy}{dx} = - 3x^2 + 3 \cdot \frac{dy}{dx}$

$\Rightarrow x \frac{d^2y}{dx^2} + \frac{dy}{dx} + 3x^2 - 3 \frac{dy}{dx} = 0$

$\Rightarrow x \frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + 3x^2 = 0$

(proved)

Ques No: 6 $If\ y = a\ cos(log\ x) + b\ sin\ (log\ x)\ then\ show\ that\ \ x^2 y_2 + x y_1 + y = 0$

sol: given that
y= a cos (log x) + b sin (log x) ……….1

Differentiating with respect to x we get

$Y_1 = - a\ sin\ (logx) \frac{d}{dx} (logx) + b\ cosx(logx) \frac{d}{dx} (logx)$

$\Rightarrow y_1 = - a\ sin(logx) \frac{1}{x} + b\ cos(logx) \frac{1}{x}$

$\Rightarrow x y_1 = - a\ sin(logx) + b\ cos(logx)$

Again Differentiating with respect x we get

$xy_2 + y_1 =- a\, cos(logx)\frac{d}{dx}(logx) - b\, sin(logx)\frac{d}{dx}(logx)$

$\Rightarrow xy_2 + y_1 =- a\, cos(logx)\frac{1}{x} - b\, sin(logx)\frac{1}{x}$

$\Rightarrow x^2y_2 + xy_1 =- cos(logx) - b\, sin(logx)$

$\Rightarrow x^2y_2 + xy_1 =- \{a\, cos(logx) + b\, sin(logx)\}$

$\Rightarrow x_2^{2y} + xy_1 =- y$……………….[from 1]

$x^2y_2 + xy_1 + y = 0$ (shown)

Ques No:7 $If = e^x\, cos\, 2x\; then\; show\; that \frac{d^{2y}}{dx^2} - 2\frac{dy}{dx} + 5y = 0$

Solution:

Let, y=$e^x cos2x$

differentiating with respect x we get

$$\frac{dy}{dx} = e^x\frac{d}{dx}(cos2x) + (cos2x)\frac{d}{dx}e^x$$

$$\Rightarrow \frac{dy}{dx} = e^x - sine2x\frac{d}{dx}(2x) - cos2xe^x$$

$$\Rightarrow \frac{dy}{dx} =- 2e^x sin2x + e^x cos2x$$

Again differentiating with x we get,

$$\Rightarrow \frac{d^{2y}}{dx^2} =- 2[e^x\frac{d}{dx}sin2x + sin2x\frac{d}{dx}e^x] + [e^x\frac{d}{dx}cos2x + cos2x\frac{d}{dx}e^x]$$

$$=- 2[e^x cos2x\frac{d}{dx}(2x) + e^x sin2x] + [e^x(- sin2x)\frac{d}{dx}2x + e^x cos2x]$$

$$= - 4e^x cos2x - 2e^x sin2x - 2e^x sin2x + e^x cos2x$$

$$= -3e^{x}cos2x - 4e^{x}sin2x$$

Now, left side

$$\frac{d^{2y}}{dx^2} - 2\frac{dy}{dx} + 5y$$

$$= -3e^{x}cos2x - 4e^{x}sin2x - 2\left(-2e^{x}sin2x + e^{x}cos2x\right) + 5e^{x}cos2x$$

$$= -3e^{x}cos2x - 4e^{x}sin2x + 4e^{x}sin2x - 2e^{x}cos2x + 5e^{x}cos2x$$

$$= 0$$

(showed)

Ques No 8:

if $x = \frac{1-t}{1+t}$ ***and*** $y = \frac{2t}{1+t}$ ***then find*** $\frac{d^2y}{dx^2}$

Sol:Differentioting we respect x we get

$$x = \frac{1-t}{1+t}$$

$$\Rightarrow \frac{dx}{dt} = \frac{1 + t\frac{d}{dt}(1-t) - (1-t)\frac{d}{dt}(1+t)}{\left(1+t^2\right)}$$

$$\Rightarrow \frac{dx}{dt} = \frac{1 + t(-1) - (1-t)\frac{d}{dt}(1+t)}{\left(1+t^2\right)}$$

$$\Rightarrow \frac{dx}{dt} = \frac{1 + t(-1) - (1-t)\frac{d}{dt}(1+t)}{\left(1+t^2\right)}$$

$$\Rightarrow \frac{dx}{dt} = \frac{1 + t(-1) - (1+t).1}{\left(1+t^2\right)}$$

$$\Rightarrow \frac{(1-t-1-t)}{\left(1+t^2\right)}$$

$y = \frac{2t}{(1+t^2)}$

Again differentiating with respect y we get

$\Rightarrow \frac{dy}{dt} = \frac{(1+t)\frac{d}{dt}(2t) - (2t)\frac{d}{dt}(1+t)}{(1+t^2)}$

$\Rightarrow \frac{dy}{dt} = \frac{(1+t).t - (2t.1)}{(1+t^2)}$

$\Rightarrow \frac{dy}{dt} = \frac{(2+2t-2t)}{(1+t^2)}$

$\Rightarrow \frac{dy}{dt} = \frac{(2)}{(1+t^2)}$

Now,

$\frac{dy}{dx} = \frac{(\frac{dy}{dt})}{(\frac{dx}{dt})}$

$\Rightarrow \frac{dy}{dx} = \frac{\frac{(2)}{(1+t^2)}}{\frac{(-2)}{(1+t^2)}}$

$\Rightarrow \frac{dy}{dx} = (-1)$

$\Rightarrow \frac{(d^2y)}{(dx^2)} = 0$

(Solved)